On the Apparent $\infty \cdot 0$ Indeterminacy of Relativistic Mass and Length Contraction: A Dimensional and Covariant Analysis of the $v \to c$ Boundary

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Abstract

Heuristic arguments in special relativity often combine the growth of "relativistic mass" $m_{\mathrm{rel}} = \gamma m$ with the Lorentz contraction $L = L_0/\gamma$ and conclude that the limit $v \to c$ produces an indeterminate form of type $\infty \cdot 0$, or that it equates a divergent mass–time product with a divergent length. We examine the schematic boundary statement $X = \lim_{v\to c}(\infty\, M \cdot T) = \infty\,(xL)$ and show that (i) it admits no dimensionally consistent left-hand side X in SI, natural ($\hbar = c = 1$), or geometrized ($G = c = 1$) units; (ii) the only monomial in G, c, $\hbar$ that could restore consistency is $\kappa = \sqrt{Gc/\hbar} = c/m_P$; and (iii) the alleged indeterminacy is absent, since $m_{\mathrm{rel}}L \equiv mL_0$ identically for all $v < c$. A covariant treatment based on the stress–energy tensor of pressureless dust confirms which quantities diverge (energy, momentum, mass per unit length, dilated time) and which remain finite and Lorentz-invariant. A γ-power-counting rule classifies all such products. We close by giving a well-posed boundary relation with genuine dimension of length that diverges in the limit, the laboratory distance covered in a fixed proper time, $\Delta x = p\tau_0/m = c\tau_0 \sinh\eta$.

1 Introduction

The Lorentz factor

$$\gamma(v) = \left(1 - \frac{v^2}{c^2}\right)^{-1/2}, \qquad 0 \le v < c, \tag{1}$$

diverges as $v \to c^-$. Several textbook kinematic effects scale as positive or negative powers of γ: the energy $E = \gamma mc^2$, the contracted length $L = L_0/\gamma$ of a rod measured along its direction of motion, and the dilated coordinate time $\Delta t = \gamma\tau_0$ corresponding to a proper interval τ_0 [1, 2, 3]. Products of such quantities are sometimes presented as paradoxical: one factor grows without bound while another vanishes, and the product is then described as an indeterminate form whose value at the light-speed "boundary" is assumed to be something non-trivial.

In this note we analyse a representative statement of this kind,

$$X \;=\; \lim_{v\to c}(\infty\, M \cdot T) \;=\; \infty\,(xL), \tag{2}$$

where M is a mass, T a time, L a length and x a dimensionless number. We ask three questions: what physical quantity X could stand on the left; whether the product of relativistic mass and contracted length is genuinely indeterminate; and what covariant statement survives once the heuristics are removed. Throughout, m denotes the invariant (rest) mass. We use the term "relativistic mass" $m_{\mathrm{rel}} \equiv E/c^2 = \gamma m$ only to engage with the heuristic. Its use is discouraged in modern treatments [4, 2].

2 Dimensional analysis

Writing $[\,\cdot\,]$ for the dimension of a quantity in the base $\{\mathsf{M}, \mathsf{L}, \mathsf{T}\}$ [5], the two sides of Equation (2) carry

$$[\,M\,T\,] = \mathsf{MT}, \qquad [\,x\,L\,] = \mathsf{L}. \tag{3}$$

The symbol ∞ denotes a limiting process and carries no dimension. Consistency would require $\mathsf{MT} = \mathsf{L}$. Table 1 shows that this fails in all common systems of units.

Table 1: Dimension of $M \cdot T$ under common unit conventions. In no case does it reduce to L.

Convention	$[\,M\,]$	$[\,T\,]$	$[\,MT\,]$	$= \mathsf{L}$?
SI	M	T	MT	no
$c = 1$	M	L	ML	no
$\hbar = c = 1$ (natural)	L^{-1}	L	1	no
$G = c = 1$ (geometrized)	L	L	L^2	no

Proposition 1 (No admissible left-hand side)**.** *There is no physical quantity X such that $[\,X\,] = [\,MT\,]$ and $[\,X\,] = [\,xL\,]$ simultaneously in any of the conventions of Table 1. Hence Equation (2), read literally, is not a valid physical equation.*

Proof. Immediate from Equation (3) and Table 1: in each row $[\,MT\,] \neq \mathsf{L}$. □

Consistency can be restored only by multiplying the left-hand side by a constant κ with $[\,\kappa\,] = \mathsf{L}\,\mathsf{M}^{-1}\mathsf{T}^{-1}$.

Proposition 2 (Unique fundamental rescaling)**.** *Among monomials $G^a c^b \hbar^d$, the unique one with dimension $\mathsf{L}\,\mathsf{M}^{-1}\mathsf{T}^{-1}$ is*

$$\kappa = \sqrt{\frac{Gc}{\hbar}} = \frac{c}{m_P} \approx 1.378 \times 10^{16}\,\mathrm{m\,kg^{-1}\,s^{-1}}, \qquad m_P \equiv \sqrt{\frac{\hbar c}{G}}. \tag{4}$$

Proof. With $[\,G\,] = \mathsf{L}^3\mathsf{M}^{-1}\mathsf{T}^{-2}$, $[\,c\,] = \mathsf{L}\mathsf{T}^{-1}$ and $[\,\hbar\,] = \mathsf{M}\mathsf{L}^2\mathsf{T}^{-1}$, matching exponents gives the linear system $3a + b + 2d = 1$, $-a + d = -1$, $-2a - b - d = -1$. Its unique solution is $a = b = \frac{1}{2}$, $d = -\frac{1}{2}$. □

The rescaled product $\kappa MT = (M/m_P)\,cT$ is a light-travel distance weighted by mass in Planck units. It is dimensionally sound, but it is a construction, not an established physical law. It is recorded here only to show what Equation (2) would minimally require.

3 The product of relativistic mass and contracted length

Let a rod of rest mass m and rest length L_0 move with speed v along its length. In the laboratory frame,

$$m_{\mathrm{rel}}(v) = \gamma(v)\,m \xrightarrow[v\to c^-]{} \infty, \qquad L(v) = \frac{L_0}{\gamma(v)} \xrightarrow[v\to c^-]{} 0. \tag{5}$$

Proposition 3 (No indeterminacy)**.** *For every $0 \le v < c$,*

$$m_{\mathrm{rel}}(v)\,L(v) = mL_0, \qquad \textit{hence} \qquad \lim_{v\to c^-} m_{\mathrm{rel}} L = mL_0. \tag{6}$$

Proof. $m_{\mathrm{rel}}L = (\gamma m)(L_0/\gamma) = mL_0$ pointwise on $[0, c)$, because $\gamma \ge 1 > 0$. A constant function has that constant as its limit. □

Remark 1. The phrase "indeterminate form $\infty \cdot 0$" describes the limits of the individual factors. It says nothing about their product, which must be found from the functions themselves. Here the product is constant before the limit is taken, so the indeterminacy is only apparent. Since m and L_0 are both frame-independent, mL_0 is a Lorentz scalar.

3.1 Which time?

The symbol T in Equation (2) is ambiguous, and the two readings give opposite results:

$$\text{proper time, } \Delta\tau = \Delta t/\gamma: \qquad m_{\mathrm{rel}}\,\Delta\tau = m\,\Delta t \quad \text{(finite for fixed } \Delta t\text{)}, \tag{7}$$

$$\text{dilated time, } \Delta t = \gamma\tau_0: \qquad m_{\mathrm{rel}}\,\Delta t = \gamma^2 m\tau_0 \xrightarrow[v\to c^-]{} \infty. \tag{8}$$

Only the second reading produces a divergence, and its dimension is MT, not L, in agreement with Proposition 1.

3.2 A power-counting rule

Each kinematic quantity considered here behaves as $Q \propto \gamma^{n_Q}$ at fixed rest-frame data (note that $\gamma\beta \to \gamma$ as $\beta \to 1$). A product $Q_1Q_2 \propto \gamma^{n_1+n_2}$ therefore diverges, stays finite, or vanishes according to whether $n_1 + n_2$ is positive, zero, or negative. Table 2 lists the exponents.

Table 2: Leading γ-exponents of laboratory-frame quantities at fixed rest-frame data ($\beta = v/c$).

Quantity	Expression	n_Q
Energy / relativistic mass	$E = \gamma mc^2,\ \ m_{\mathrm{rel}} = \gamma m$	$+1$
Momentum	$p = \gamma mv$	$+1$
Dilated time	$\Delta t = \gamma\tau_0$	$+1$
Number density	$n = \gamma n_0$	$+1$
Mass per unit length	$\lambda = \gamma^2 m/L_0$	$+2$
Contracted length	$L = L_0/\gamma$	-1
Proper time (fixed Δt)	$\Delta\tau = \Delta t/\gamma$	-1

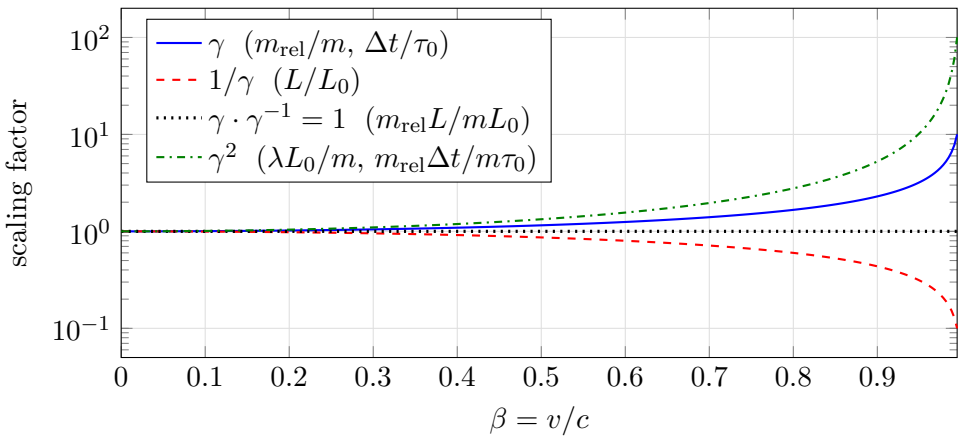


Figure 1: Scaling of laboratory-frame quantities with β. The product of relativistic mass and contracted length (dotted) is identically constant. Products with net positive γ-exponent diverge as $\beta \to 1$.

4 Covariant formulation

The heuristic of Section 3 can be replaced by a manifestly covariant derivation. We use the metric signature $(+,-,-,-)$. Model the rod as pressureless dust of rest density ρ_0 occupying, in its rest frame, a cylinder of cross-section A and length L_0, so that $m = \rho_0 AL_0$. Its stress–energy tensor and four-velocity are [6, 3]

$$T^{\mu\nu} = \rho_0\, u^\mu u^\nu, \qquad u^\mu = \gamma\,(c,\, v,\, 0,\, 0), \qquad u_\mu u^\mu = c^2. \tag{9}$$

For uniform motion $\partial_\mu T^{\mu\nu} = 0$. By von Laue's theorem the integral of $T^{\mu 0}$ over a hypersurface $t = \text{const}$ is therefore a four-vector [3, 7]. The relevant components are

$$T^{00} = \gamma^2 \rho_0 c^2, \qquad T^{10} = \gamma^2 \rho_0 c\, v. \tag{10}$$

The factor γ^2 has two independent origins: one factor of γ comes from the energy per particle, and the other from the increase of number density under contraction. Integrating over the contracted laboratory volume $V = AL_0/\gamma$ gives

$$P^\mu = \frac{1}{c}\int_V T^{\mu 0}\, \mathrm{d}^3x = \frac{1}{c}\,\gamma^2\rho_0\,(c^2,\, cv,\, 0,\, 0)\,\frac{AL_0}{\gamma} = \gamma m\,(c,\, v,\, 0,\, 0), \tag{11}$$

that is, $E = cP^0 = \gamma mc^2$ and $p = P^1 = \gamma mv$. The invariant norm is

$$P_\mu P^\mu = \gamma^2 m^2 (c^2 - v^2) = m^2 c^2, \tag{12}$$

independent of v. The rest mass does not diverge.

From Equation (11) the divergent and finite quantities follow directly:

$$\text{divergent:} \quad E = \gamma mc^2, \quad p = \gamma mv, \quad \lambda \equiv \frac{E/c^2}{L} = \frac{\gamma^2 m}{L_0}, \tag{13}$$

$$\text{finite:} \quad \frac{E}{c^2}\,L = mL_0, \qquad P_\mu P^\mu = m^2 c^2. \tag{14}$$

This reproduces Proposition 3 from the tensor without using the notion of relativistic mass at any stage.

5 A well-posed boundary relation

If the intent behind Equation (2) is a relation that has dimension L and diverges at the boundary, standard kinematics supplies one. Introduce the rapidity η through

$$\beta = \tanh\eta, \qquad \gamma = \cosh\eta, \qquad \gamma\beta = \sinh\eta, \tag{15}$$

so that $v \to c^-$ corresponds to $\eta \to \infty$. The rapidity maps the bounded interval $[0, c)$ onto $[0, \infty)$, which removes any sense of a finite "boundary".

Proposition 4 (Divergent laboratory displacement)**.** *The laboratory-frame distance covered by a body during a fixed proper time τ_0 is*

$$\Delta x = v\,\gamma\tau_0 = \frac{m_{\mathrm{rel}}}{m}\,v\,\tau_0 = \frac{p\,\tau_0}{m} = c\tau_0 \sinh\eta \;\sim\; \frac{c\tau_0}{2}\,e^\eta \xrightarrow[\eta\to\infty]{} \infty. \tag{16}$$

Proof. $\Delta x = v\Delta t$ with $\Delta t = \gamma\tau_0$. The remaining equalities follow from $p = \gamma mv$ and Equation (15). The asymptotic form uses $\sinh\eta = \frac{1}{2}(e^\eta - e^{-\eta})$. □

Equation (16) is the closest correct analogue of Equation (2). A divergent mass factor (m_{rel}, or equivalently p) multiplies a time (τ_0), and the factor v/m, of dimension $\mathsf{L}\,\mathsf{M}^{-1}\mathsf{T}^{-1}$, converts the product into a length. It plays exactly the role that Proposition 2 assigned to the universal constant c/m_P, but with the body's own velocity and rest mass in place of c and m_P.

6 Physical inaccessibility of the limit

For a massive body the limit $v \to c$ is a statement about asymptotic behaviour, not about a reachable state. The work required to accelerate the body from rest is

$$W = (\gamma - 1)\, mc^2 \xrightarrow[v \to c^-]{} \infty. \tag{17}$$

All divergences in Equations (8), (13) and (16) are therefore properties of a one-parameter family of physically realizable states with $v < c$. None of them describes a state at $v = c$. Massless particles, for which $v = c$ exactly, are described by null four-momenta with $P_\mu P^\mu = 0$, and the quantities γ, m_{rel} and L_0/γ are undefined for them. (The visual appearance of rapidly moving bodies differs again from the contracted length because of light-travel-time effects [8, 9]. That issue is separate from the measured length considered here.)

7 Conclusions

1. The schematic boundary equation $X = \lim_{v \to c}(\infty\, M \cdot T) = \infty\,(xL)$ has no dimensionally consistent left-hand side (Proposition 1). The only fundamental-constant rescaling that repairs it is $\kappa = \sqrt{Gc/\hbar} = c/m_P$ (Proposition 2).

2. The product of relativistic mass and contracted length is not indeterminate. It equals the Lorentz scalar mL_0 at every sub-luminal speed (Proposition 3). The same result follows covariantly from $T^{\mu\nu} = \rho_0 u^\mu u^\nu$ (Equation (14)).

3. Whether a product of kinematic quantities diverges is fixed by its net γ-exponent (Table 2). Energy, momentum, mass per unit length and dilated time diverge; mL_0 and $P_\mu P^\mu$ do not.

4. A well-posed relation of dimension L that diverges at the boundary is $\Delta x = p\tau_0/m = c\tau_0 \sinh\eta$ (Proposition 4).

A Asymptotics near $\beta \to 1$

Let $\varepsilon \equiv 1 - \beta \to 0^+$. Then $1 - \beta^2 = \varepsilon(2 - \varepsilon)$ and

$$\gamma = \frac{1}{\sqrt{2\varepsilon}}\left(1 + \frac{\varepsilon}{4} + O(\varepsilon^2)\right), \qquad \frac{L}{L_0} = \sqrt{2\varepsilon}\left(1 - \frac{\varepsilon}{4} + O(\varepsilon^2)\right). \tag{18}$$

The leading singular terms cancel exactly in $m_{\mathrm{rel}}L/(mL_0)$, consistent with Proposition 3, which holds to all orders. In terms of rapidity, $\varepsilon = 1 - \tanh\eta = 2e^{-2\eta} + O(e^{-4\eta})$, so $\gamma \simeq \frac{1}{2}e^{\eta}$, in agreement with Equation (16).

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